

9. A force of magnitude 20N is applied to a block of mass 2kg that lies on a rough, horizontal surface as shown in figure below. The coefficient of kinetic friction between the block and surface is 0.4. What is the magnitude of the acceleration of the block?

- A. 10 m/s^2
B. 5.2 m/s^2
C. 4 m/s^2
D. 2.8 m/s^2

20. Two solid spheres made of gold and silver having equal masses are suspended by strings and fully submerged in a tank of water. Given that the density of gold is greater than that of silver, which one of the following statement is true? (1)
- A. The buoyant force on each sphere is the same.
 - B. The volume of water displaced by the gold sphere is larger than that of the silver sphere.
 - C. The tension in the string supporting each sphere is the same.
 - D. The buoyant force on the silver sphere is greater than that on the gold sphere.
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21. The cross section areas of the small and large pistons of a hydraulic lift shown in figure are 4.0 cm^2 and 0.02 m^2 , respectively. What is the magnitude of the force that must be applied to the small piston to lift a weight $w = 15,000 \text{ N}$ placed on the larger piston? (1)
- [Figure: A U-shaped vessel drawn in cross section with thick walls, holding a fluid. Its left limb is narrow and its right limb is wide; each limb is closed by a piston. A vertical arrow pointing downward is drawn inside the left limb, above the small piston, and is labelled F_1 . A shaded rectangular block labelled W rests on top of the large piston in the right limb, which sits higher than the small piston.]
- A. $3 \times 10^2 \text{ N}$
 - B. $2 \times 10^4 \text{ N}$
 - C. $3 \times 10^5 \text{ N}$
 - D. $3 \times 10^6 \text{ N}$
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22. An incompressible fluid flows through a horizontal pipe whose diameter varies along its length. The sum of the kinetic energy per unit volume and pressure of the fluid at different section of the pipe would_____ (1)
- A. depend on the length of the pipe.
 - B. Increase as the pipe diameter increases.
 - C. Increase as the pipe diameter decreases.
 - D. Remain the same as the pipe diameter changes.
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23. The amount of heat required to raise the temperature of an object of mass $m = 1 \text{ Kg}$ by 1 Kelvin is known as_____ (1)
- A. specific heat capacity
 - B. heat capacity
 - C. expansion coefficient
 - D. internal energy
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24. The first law of thermodynamics takes the form $dU = dW$. This equation is valid if the process is_____ (1)
- A. isobaric
 - B. isochoric
 - C. isothermal
 - D. adiabatic
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25. An ideal gas undergoes an isothermal process starting with a pressure of $2 \times 10^5 \text{ pa}$ and a volume of 6 cm^3 . Which one of the following might be the pressure and volume of the final state? (1)
- A. $4 \times 10^5 \text{ pa}$ and 4 cm^3
 - B. $4 \times 10^5 \text{ pa}$ and 3 cm^3
 - C. $5 \times 10^5 \text{ pa}$ and 3 cm^3
 - D. 10^5 pa and 10 cm^3
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26. The position of a particle moving with simple harmonic motion is given by an expression $x = A \cos(\omega t + \Phi)$, where A is the amplitude of the oscillation, t is time, ω is angular frequency, and Φ is the phase constant. If $\Phi = \pi/2$, what will be the sign of the particle's position and velocity at $t = 7T/8$, where T is the period of oscillation? (1)
- A. The position is positive, and the velocity is negative.
 - B. The position and velocity are both positive.
 - C. The position and velocity are both negative.
 - D. The position is negative and the velocity is positive.

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27. A given minimum point (trough) of a sinusoidal wave is traveling 480 cm long a rope in 12s. If the oscillator that generates the wave completes 48 vibrations in 30s, what is the wave length of the wave? (1)
- A. 25m
B. 15m
C. 0.25m
D. 0.15m
-
28. A wave traveling from a medium of index of refraction n_1 toward a medium of index of refraction n_2 undergoes partly transmission and partly reflection. If n_2 is greater than n_1 which one of the following statements is not correct? (1)
- A. The speed of the wave in the medium with reflective index n_1 is smaller than with n_2
B. The incident and the reflected waves are in phase.
C. The angle of reflection is the same as the angle of incidence.
D. The reflection angle is less than the incidence angle.
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29. Which one of the following statement is correct? (1)
- A. The electric field strength due to two point charges of equal magnitude and the same sign is zero only at one point on the line joining the two charges.
B. If a point charge of mass m is released from the rest in uniform electric field, it will move along a line of force with uniform speed.
C. The variation of the strength of the electric field due to an electric dipole is inversely proportional with the square of the distance from its center of mass.
D. An electric dipole moment p in a uniform external electric field E is in stable equilibrium if p and E are in opposite directions.
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30. In a certain region of space, a uniform electric field is directed in the positive x direction. If a particle with negative charge is carried from $x = 20\text{cm}$ to $x = 60\text{cm}$, what can we conclude about the electric potential and the electric potential energy of the charge-field system? (1)
- A. The electric potential at $x = 20\text{cm}$ is larger than that at $x = 60\text{cm}$.
B. The electric potential at $x = 20\text{cm}$ is the same as that at $x = 60\text{cm}$.
C. The electric potential energy decrease.
D. The electric potential energy increase.
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31. A parallel plate capacitor has a capacitance of $10\ \mu\text{F}$. If the distance between the plates is doubled, keeping all other quantities constant, what is the new capacitance of the capacitor? (1)
- A. $0.2\ \mu\text{F}$
B. $5\ \mu\text{F}$
C. $10\ \mu\text{F}$
D. $20\ \mu\text{F}$
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32. If several resistors are connected in parallel, the equivalent resistance is (1)
- A. greater than any of the resistances in the group.
B. is equal to the average resistances of the group.
C. less than any of the resistances in the group.
D. equal to the sum of the resistances in the group.
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33. In aluminium and copper wires with circular cross sections have equal length and equal resistance, then the ratio of the radius of aluminium to that of copper should be the same as that of the_____ (1)
- A. ratio of the resistivity of aluminium to copper
B. ratio of the resistivity of copper to aluminium
C. square root of the ratio of the resistivity of aluminium to copper
D. square root of the ratio of the resistivity of copper to aluminium

34. In the circuit shown below, what is the potential difference between point b and a? (1)
- [Figure: A circuit made of three horizontal branches joined between a left-hand vertical wire and a right-hand vertical wire. Point a is marked as a node on the left wire where the middle branch meets it; point b is marked as a node on the right wire where the middle branch meets it. Top branch: a resistor labelled $16\ \Omega$, with a current arrow drawn just beneath it pointing to the LEFT and labelled I_3 . Middle branch, running from a to b: a resistor labelled $12\ \Omega$, then a battery labelled 24 V , with a current arrow drawn beneath the resistor pointing to the RIGHT and labelled I_2 . Bottom branch: a battery labelled 12 V on the left and a resistor labelled 9 on the right, with a current arrow drawn beneath the resistor pointing to the LEFT and labelled I_1 .]
- A. 19.2V B. 24.0V
C. 13.0V D. 12.0V
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35. An electron moving at a speed of $1.5 \times 10^7\text{ m/s}$ enters a region of uniform magnetic field of magnitude $2 \times 10^{-3}\text{ T}$, direction perpendicular to the field. The path followed by the electron is circular. What is the radius of the circular path? (1)
- A. 0.43cm B. 4.3cm
C. 43cm D. 430cm
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36. If a wire carrying current in the plane of this paper toward the top of the page experiences a magnetic force toward the right edge of the page, what will be the direction of the magnetic field causing this force? (1)
- A. In the plane of the page and toward the left edge.
B. In the plane of the page and toward the bottom edge.
C. Perpendicular and directed out of the plane of the page.
D. Perpendicular and directed in to the plane of the page.
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37. A square flat loop of wire is placed in a uniform magnetic field that is in the y direction. The magnetic flux through the loop is a maximum if the plane of the loop is in the (1)
- A. xy plane B. xz plane
C. yz plane D. inclined 45° with respect to the xy plane
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38. Which one of the following actions increases the magnetic energy density in the interior of a very long current carrying solenoid? (1)
- A. Increasing the number of turns per unit length of the solenoid.
B. Increasing the cross section area of the solenoid.
C. Increasing the length of the solenoid keeping the number of turns per unit length fixed.
D. Decreasing the cross section area of the solenoid.
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39. Figure below shows a rectangular loop of wire of width L and resistance R. one end of the loop is in a uniform magnetic field of strength B directed perpendicular in to the plane of the page, the loop is pulled to the right at a constant speed v. what are the magnitude and direction of the induced current in the loop? (1)
- [Figure: On the left, a shaded rectangular region filled with four rows of X symbols, representing a uniform magnetic field directed into the plane of the page. A rectangular loop of wire is drawn across the figure so that its left-hand portion lies inside the field region (its left side is buried among the X symbols) and its right-hand side lies outside the field region, clear of the X symbols. A short double-headed vertical arrow beside the loop's right side marks the loop's width and is labelled L. A horizontal arrow points to the right from the loop's right side and is labelled v.]
- A. $I = BLvR$ and clockwise B. $I = BLv/R$ and clockwise
C. $I = BLvR$ and counter clockwise D. $I = BLv/R$ and counter clockwise

- 40.** A series RLC circuit has a total impedance of $75.0\ \Omega$. If the resistance is $R = 45.0\ \Omega$ and the average power delivered to the circuit is $P = 352.8\ \text{W}$, what is the root mean square value of the ac voltage source? (1)
- A. 83V
B. 116V
C. 210V
D. 297V
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- 41.** Which of the following phenomena most clearly demonstrates the wave nature of electrons? (1)
- A. The photoelectric effect
B. Blackbody radiation
C. The Compton effect
D. Diffraction of electrons by crystals
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- 42.** When the radioactive element $^{144}\text{Nd}_{60}$ decays to $^{140}\text{Ce}_{58}$, which one of the following particles is emitted? (1)
- A. A beta particle
B. An alpha particle
C. a proton
D. a gamma ray

— END OF EXAMINATION —

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